

Market Structure and Development: Firm Competition in Jamaica's Hotel Industry

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Development Workshop
03/26/2026

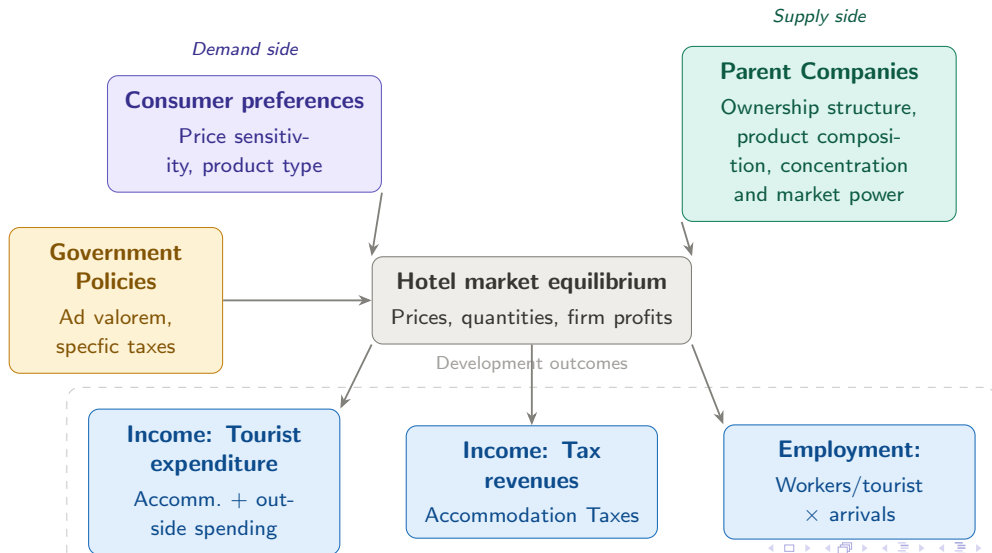
Motivation

- Growing specialization in services is a defining feature of many emerging economies; tourism has been a major driver.
- Existing work has discussed:
 - The ability of tourism to contribute to economic development through spillovers to other sectors.(McKetty 2026; Faber and Gaubert 2019; Wattanakuljarus and Coxhead 2006)
 - The impact of tourism in urban settings like Barcelona and Amsterdam.(Almagro and Domínguez-lino 2025; Allen et al. 2021)
- What has not been discussed is the role of **market structure**, **consumer preferences**, and **government policy** on **local development outcomes**.
- This paper fills these gaps using Jamaica's hotel industry as a laboratory.

Research Questions

- ① What is the price-sensitivity of tourists choosing between Jamaica's different accommodation types?
- ② How do ownership structure and product composition shape three development outcomes:
 - ① Total tourist expenditures (accommodation + non-accommodation)
 - ② Government tax revenues
 - ③ Accommodation employment
- ③ What policy interventions best align market outcomes with development goals?

Conceptual Framework



What I Do

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- ① **Estimate consumer demand** — nested logit model using over 20 years of exit survey data (109,708 tourist parties).
- ② **Estimate firm supply** — profit-maximizing parent companies in Nash-Bertrand competition, estimated via GMM.
- ③ **Run counterfactual simulations:**
 - ① The “Spanish Invasion” of Montego Bay (multinational entry)
 - ② Ownership structure in Negril (multi-property vs. single-property)
 - ③ Room Tax (GART) reform
 - ④ General Consumption Tax (GCT) reform

Preview of Main Findings

- Demand is **price-inelastic** (mean own-price elasticity = -1.43), consistent with Jamaica's premium tourism positioning.
- All-inclusive resort guests are especially **inelastic**: multinationals entering Montego Bay largely attracted *new* visitors rather than diverting them from incumbents.
- For maximizing arrivals and employment, **entry by non-incumbent firms** outperforms expansion by incumbents.
- A more **progressive ad-valorem tax structure** (lowering GCT for small accommodations, raising it for large all-inclusives) raises tax revenues by $\approx 55\%$ with minimal impact on employment and total tourist spending.

Related Literature

- **Tourism and accommodations:** (McKetty 2026; Almagro and Domínguez-lino 2025; McClure 2025; Farronato and Fradkin 2022; Allen et al. 2021; Faber and Gaubert 2019)
- **Multinational entry and market structure:** (Atkin et al. 2018; Talamas Marcos 2025; Ghazzai et al. 2023; Medina 2024; Busso and Galiani 2019; Bergquist and Dinerstein 2020)
- **Services and development:** (Peters et al. 2026; Fan et al. 2023; Hsieh and Rossi-Hansberg 2023; Nayyar and Davies 2023)

Contribution: First structural analysis of how hotel market organization shapes development outcomes in a tourism-dependent emerging economy; integrates demand, supply, and tax incidence.

Roadmap

- 1 Introduction & Motivation
- 2 Data
- 3 The Jamaican Hotel Industry
- 4 Model
- 5 Estimation Results
- 6 Counterfactual Simulations
- 7 Conclusion

Data I: Ministry of Tourism Exit Surveys

- Surveys of departing tourists at both international airports, conducted annually 2001–2023 (excluding 2007 and 2015).
- **Final sample:** 109,708 tourist parties; 222,292 total travellers.
- Key variables:
 - Accommodation type and name
 - Length of stay and party size
 - Total accommodation expenditure; itemized non-accommodation spending (food, transport, entertainment, shopping)
 - Origin country, purpose of visit, demographics
- Enables construction of *per-person per-booking* prices for each accommodation product in each market-year.

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The number of persons covered by these expenses _____

A. **Accommodation** (give total bill inclusive of meals, etc.

How much of this was for _____ Room

Food and Bev _____

B. **Food and Beverage outside Accommodation**

• Groceries _____

Restaurant _____

Nightclub _____

C. **Entertainment**

i) Visitor attractions (e.g. Rafting, Tours, etc.) _____

ii) Sports & Recreation (e.g. Golf Fees, Scuba Diving et

D. **Transportation**

Taxi _____

Car Rental _____

MOT Exit Survey (example)

Data II: Accommodation and Arrival Data

- **MOT Annual Accommodation Reports:** number and type of active properties, room capacity by category, resort area, and year.
- **MOT Aggregate Arrival Data:** total stopover arrivals by resort area and accommodation type — used to scale exit-survey counts to market-level quantities.
- **Tax Data:** Guest Accommodation Room Tax (GART) schedules (post-2013) by property size tier; General Consumption Tax (GCT) schedules.
- **Supplementary:** Google Trends search indices; parish-level property transaction values (2004+) for instruments.

Sample Summary

Table: Resort Area × Accommodation Category Arrivals (2001–2023)

Resort Area	Category	Metric	2001	2023	% Δ
Montego Bay	Large All-Inclusive	Room Nights	780,971.00	2,139,890.00	174.00
		Parties	149,239.00	509,214.00	241.00
		Persons	285,154.00	1,088,590.00	282.00
	Medium European Plan	Room Nights	60,941.00	0.00	0.00
		Parties	14,217.00	0.00	0.00
		Persons	26,786.00	0.00	0.00
	Small European Plan	Room Nights	140,606.00	96,863.00	−31.00
		Parties	30,002.00	20,870.00	−30.00
		Persons	56,865.00	44,757.00	−21.00

Jamaica: Economic Background

- Upper-middle-income island nation; GDP per-capita $\approx$ USD 5,186 (2021, World Bank).
- Services $\approx$ 58% of economic activity.
- Tourism policy dates to independence (1962):
 - Hotels Incentive Act (1971)
 - Tourist Board Act (1955)
 - Capital allowances, tariff relief, employment credits
- Primary tourism product: “Sun, Sand, and Sea.”



Jamaica

Arriving Tourist Characteristics (2001–2023)

	Mean	Median	S.D.
Share of Bookings: All-Inclusive Plan	0.84	1.00	0.36
Mean Total Exp. per Person	1,885.44	1,711.60	1,020.42
Accommodation Share	0.61	0.62	0.14
Total Outside Exp. Share	0.39	0.38	0.14
Share Reservations Montego Bay	0.45	0.00	0.50
Share Visiting for Vacation	0.91	1.00	0.29
Share 7 Days or Under	0.80	1.00	0.40
Share 8 Days or Over Stay	0.20	0.00	0.40
Share of Spanish-Owned AI Brands	0.33	0.00	0.47
Share US/International-Owned AI Brands	0.31	0.00	0.46
Booking Observations	109,708.00		
Individual Observations	222,292.00		

Table: Summary Statistics: Exit Survey Sample

The Six Resort Areas

- The Ministry of Tourism divides Jamaica into **6 Resort Areas** (RAs) for marketing and policy.
- Each RA provides a distinct vacation experience.
- RAs: **Kingston, Montego Bay, Ocho Rios, Negril, Port Antonio, Southcoast/Mandeville.**
- My market definition follows these RAs.
- A *market* = all tourism in one RA for a 12-month window (Nov–Oct).

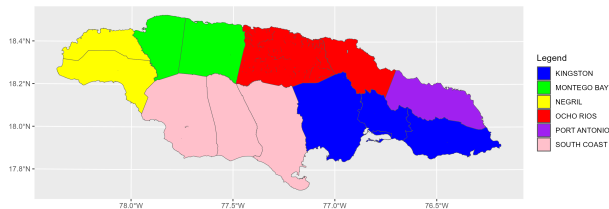


Figure: Jamaican Resort Areas

Five Accommodation Types

Jamaica's hotels are classified by the Ministry of Tourism into five categories that form the foundation of my demand model:

All-Inclusive (AI)

- **Large AI** — ≥ 150 rooms; all meals, entertainment, spa, specialty restaurants bundled into room rate.
- **Small AI** — < 150 rooms; same bundled concept, often boutique.

European Plan (EP)

- **Large EP** — ≥ 150 rooms; accommodation only; amenities unbundled; mainly serves business/urban travellers.
- **Medium EP** — 75–149 rooms; unbundled; mixed leisure/business.
- **Small EP** — < 75 rooms; villas, guesthouses, boutique properties; high share of outside spending.

The AI/EP distinction is the single most important product dimension for demand and spending patterns.

Product Type I: Large All-Inclusive Resorts



Figure: Sandals Montego Bay (Large AI)

- Dominated by large multinational chains: **Sandals/Beaches, Couples Resorts, Riu, Iberostar, Hyatt.**
- Guests pay a single price covering all meals, drinks, entertainment, and many excursions.
- **Low outside spending:** tourists have limited incentive to spend outside the resort.
- High brand recognition; targeted at leisure travellers from North America and Europe.
- Dominant and *growing* share of room-nights sold.

Product Type II: Large European Plan Hotels



Figure: Large EP Hotel, Kingston

- Concentrated in Kingston and Montego Bay business corridors.
- International and local brands: **Marriott**, **Wyndham**, **Courtyard**, and local independent operators.
- Room rate covers accommodation only; food and services purchased à la carte.
- Higher outside spending than AI resorts; guests are more integrated into local economy.
- Shrinking room-night share as AI sector grows.

Product Types III: Small Accommodations



(a) Resort Villa (Small EP)



(b) Guesthouse (Small EP)

- Villas, guesthouses, boutique inns; mostly owner-operated or locally owned.
- **Highest outside spending** of any category — guests eat locally, use local transport, hire local guides.
- Most affected by the rise of all-inclusives and short-term rental platforms.

Why Product Type Matters: Spending Spillovers

Table: Outside-Hotel Spending Shares by Accommodation Type

	Large AI		Large EP		Medium EP		Small AI		Small EP	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Outside Non-Attraction Share	0.31	0.12	0.30	0.18	0.34	0.22	0.33	0.14	0.39	0.19
Total Outside Exp. Share	0.39	0.12	0.33	0.18	0.36	0.22	0.40	0.14	0.43	0.20
Accommodation Share	0.61	0.12	0.67	0.18	0.64	0.22	0.60	0.14	0.57	0.20
Outside Non-Attraction Spending	567.45	350.61	473.71	548.41	568.30	628.84	535.68	381.84	619.62	536.63
Total Outside Spending	733.97	419.76	514.72	563.05	604.87	647.01	668.94	439.99	691.34	569.80
Accommodation Spending	1,202.57	710.71	1,048.67	855.38	1,182.42	1,027.71	1,033.64	665.48	998.40	859.56

Key takeaway: Small EP guests spend a larger share of their trip budget *outside* the hotel, creating stronger linkages with local restaurants, transport, and retail.

Market Concentration: Major Parent Companies

Table: Overview of Active Major Hotel Brands (All Years)

Brand	Parent Company	Entry Year	Adults Only	All-inclusive	Properties	Resort Areas
Domestic						
Couples	Couples Resorts	Pre-2000	✓	✓	4	2
Sandals	Sandals Resorts	Pre-2000	✓	✓	7	4
Beaches	Sandals Resorts	Pre-2000		✓	2	2
Spanish Multinationals						
Riu	Riu Hotels & Resorts	2001		✓	6	3
Iberostar	Iberostar Group	2007		✓	3	1
Grand Palladium	Fiesta Group	2008		✓	1	1
Grand Bahia	Grupo Pinero	2008		✓	1	1
Riu Palace	Riu Hotels & Resorts	2013	✓	✓	1	1
US/International						
Hedonism	Marshmallow Limited	Pre-2000	✓	✓	2	2
Sunset	Sunset Resorts	Pre-2000	✓	✓	2	2
Royal Decameron	Decameron Hotels & Resorts	2002		✓	2	2
Hilton	Blackstone Group	2009		✓	1	1
Azul	Karisma Hotels & Resorts	2009	✓	✓	1	1
Holiday Inn	Octagon Hotels Group	2009		✓	1	1
Jewel	Playa Hotels and Resorts	2010		✓	4	2

Concentration: Top-5 Parent Companies

Table: Top-5 Booking Shares

Resort Area	2001	2012	2023
Montego Bay	0.72	0.61	0.59
Ocho Rios	0.81	0.75	0.85
Negril	0.50	0.68	0.65
Kingston	0.54	0.41	0.36

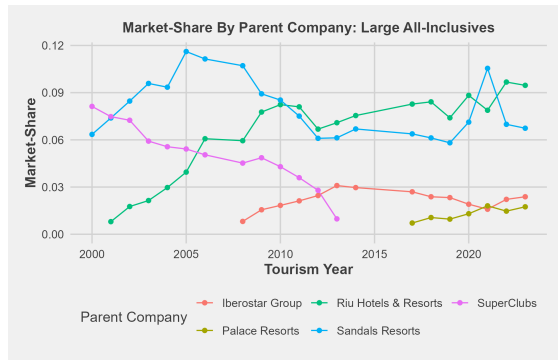


Figure: Parent Company Booking Shares

Evolution of Room Supply by Accommodation Size

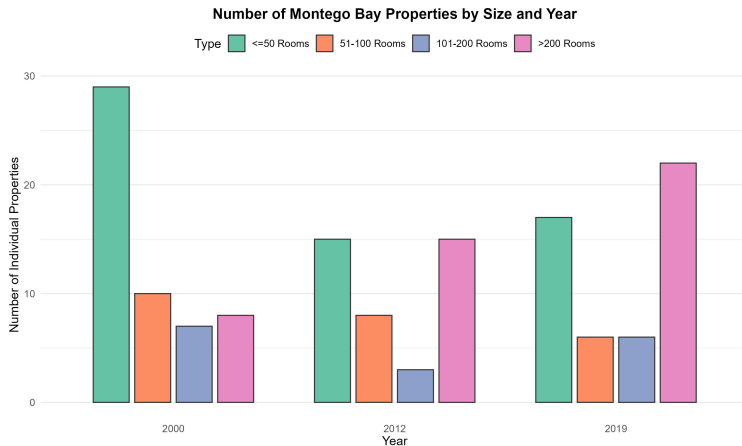


Figure: Breakdown of Properties by Accommodation Type: Montego Bay

Montego Bay: The Spanish Invasion (2006–2012)

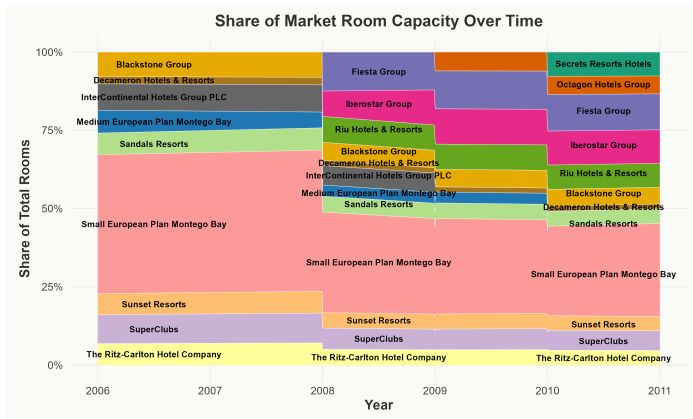


Figure: Room Capacity in Montego Bay — Entry of Spanish-Owned Multinationals

Context for Counterfactual 1: Riu Hotels, Iberostar, and Fiesta Group each opened properties in Montego Bay between 2006 and 2012, dramatically increasing Large AI capacity.

Model: Overview

Demand Side

Nested logit: tourists choose among *bookings* $b = (\text{hotel } h, \text{duration bin } d)$ in market m or an outside option. Two nests: **Large Accommodations** and **Small/Medium Accommodations**.

[► Utility Specification](#)[► Nesting Diagram](#)[► Estimating Equation](#)

Supply Side

Parent companies own portfolios of properties and set prices in Nash-Bertrand competition. Marginal costs have a *linear* component (plan- and duration-specific) plus a *nonlinear* soft capacity congestion term.

[► Profit Function](#)[► FOC / Pricing Equation](#)

Model Setup I: Markets and Products

Market $m = \text{RA} \times \text{Year}$

■ Montego Bay ■ Negril-SC
■ Kingston ■ OR-PA

4 RAs $\times$ ~ 20 yrs ≈ 80 markets

Market size $M_m = 2 \times \text{bookings}$

Slack captures tourists who could have visited but chose elsewhere

$\rightarrow$ outside-option share s_{0m}

Individual properties

Large AI



Sandals Riu

Indiv. property

≥ 150 rms, bundled

Large EP



Marriott Wyndham

Indiv. property

≥ 150 rms, unbundled

Inside option

Resort Area aggregates

Small AI



boutique

RA aggregate

< 150 rms, bundled

Medium EP



75–149 rms

RA aggregate

75–149 rms, EP

Small EP



villa / guesthouse

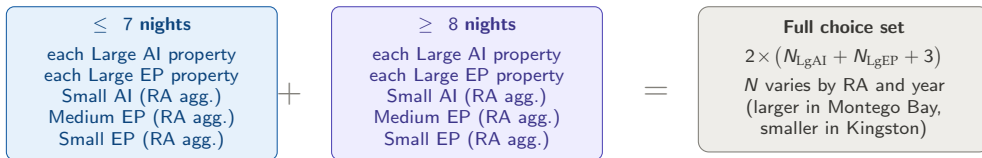
RA aggregate

< 75 rms, EP

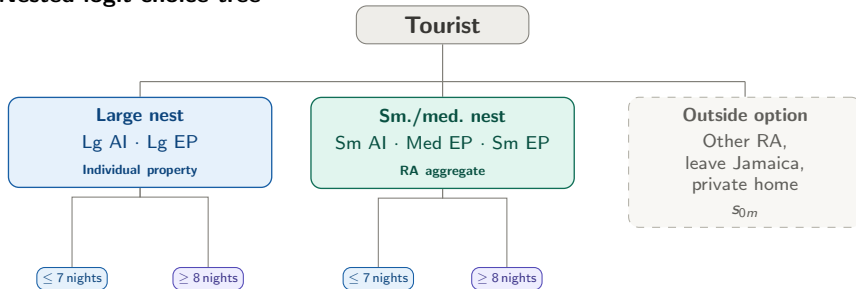
Outside option: leave Jamaica · visit a different resort area (separate market) · stay in a private home / unregistered accommodation in the RA

Model Setup II: Duration Bins and Choice Tree

Duration bins cross the entire inside-option choice set



Nested logit choice tree



Demand-Side Identification Strategy

- **Challenge:** prices are correlated with unobserved demand shocks ξ_{bm} .
- **Price Instruments used:**
 - ① *Property Capacity:* The number of rooms for a particular hotel property.
 - ② *Same-Year Portfolio Capacity in Other Resort Areas:* Total number of rooms owned by parent company in other resort areas during the same year.
 - ③ *Property Capacity x Max Property Tax:* Interaction of property capacity with the highest property tax band for a given year.
 - ④ *Same-Year Portfolio Capacity in Other Resort Areas x Max Property Tax:* Interaction of room capacity in other resort areas with the maximum national property tax band for a given year.
 - ⑤ *Same-Year Portfolio Capacity in Other Resort Areas x Minimum Wage:* Interaction of parent company room capacity in other resort areas with national minimum wage.
- **Within-Nest Share Instruments Used**
 - ① *Number of Products In The Nest:* Number of products in the nest for a given product.
 - ② *Total Number of Products In The Market:* The number of products across both nests.

Supply-Side Identification Strategy

- **Challenge:** Congestion may be correlated with unobserved shocks to parent companies.
- **Instruments For Congestion:**
 - ① *Guest Accommodation Room Tax:* The rate for the guest accommodation room tax, based on property size.
 - ② *Mean Booking Price for Different Duration, Resort Area, and Company:* Mean per-person booking price for reservation of a different length, for rival properties in other resort areas.
 - ③ *Number of Rooms Same Market-Different Category:* Aggregate room capacity in the market for properties of a different category.
 - ④ *Number of Rooms in Different Market in Different Category:* Aggregate number of rooms in other categories in other resort areas in a given year.

Demand Parameter Estimates

Table: Nested Logit Demand Parameter Estimates

Variables	(1)
α : Price	-0.010*** (0.001)
σ_{LH} : Log LH Within Nest Share	0.144*** (0.053)
σ_{SMH} : Log SMH Within Nest Share	0.236* (0.120)
F Statistic	30.5
Mean Own-Price Elasticity	-1.67
Median Own-Price Elasticity	-1.49

Notes: Standard errors in parentheses. σ_{LH} and σ_{SMH} are the within-nest correlation parameters for Large and Small/Medium accommodation nests, respectively.

Supply Parameter Estimates

Table: Accommodation Supply Estimates (GMM)

Parameter	Category	Estimate	Std. Error
Panel A: Linear Marginal Cost			
C_7	Length of Stay: ≤ 7 Days	254***	21
C_8	Length of Stay: ≥ 8 Days	480***	27
Panel B: Nonlinear Congestion Parameter (γ)			
γ_{EP}	Hotel Type: European Plan	265***	44
γ_{AI}	Hotel Type: All-Inclusive	742***	171
Model Statistics			
Number of Observations			890
λ (Dampening Parameter)			50
J Criterion			0.023

Notes: All values in 2024 USD. C_b is the linear marginal cost component for duration bin b . N = product $\times$ market observations.

Recovered Prices, Markups, and Marginal Costs

Table: Recovered Summary by Accommodation Category

LOS	Mean Price	Mean Markups	Mean Marginal Costs	Mean Own-Price Elasticity	Mean EPS per \$100	Mkts.
Large All-Inclusive						
≤ 7 Days	1,205.03 (171.30)	864.56 (27.28)	340.47 (170.88)	-1.47 (0.21)	-12.14 (0.15)	57
≥ 8 Days	1,415.03 (250.27)	860.61 (21.58)	554.41 (240.89)	-1.79 (0.32)	-12.66 (0.09)	57
Medium European Plan						
≤ 7 Days	1,664.75 (488.63)	837.98 (125.91)	826.77 (578.42)	-2.13 (0.75)	-12.59 (1.16)	22
≥ 8 Days	1,960.30 (681.40)	860.86 (131.18)	1,099.44 (751.33)	-2.76 (1.00)	-14.01 (0.36)	22
Small European Plan						
≤ 7 Days	1,061.89 (165.14)	894.40 (61.44)	167.48 (152.37)	-1.28 (0.19)	-12.12 (0.53)	30
≥ 8 Days	1,207.10 (269.72)	896.20 (63.18)	310.90 (258.63)	-1.62 (0.36)	-13.42 (0.39)	30

Notes: Per-person, per-booking in 2024 USD. Standard deviations in parentheses.

Own-Price Elasticities by Category



Figure: Mean Own-Price Elasticities by Accommodation Category and Duration Segment

Diversion Ratios: Where Do Displaced Tourists Go?

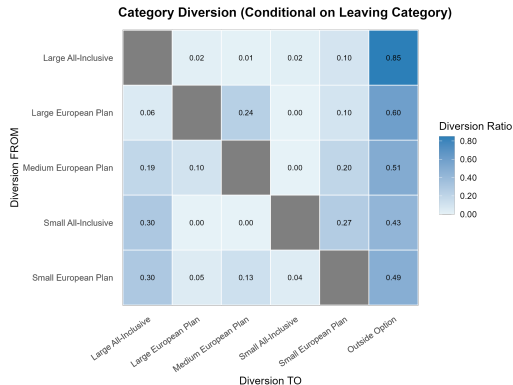


Figure: Heatmap of Mean Diversion Ratios Across Lodging Categories

Key result: Large AI tourists largely divert to the outside option (i.e. they would not visit Jamaica), confirming weak within-market substitution for the multinational brands.

Price Summary by Category and Length of Stay

Table: Mean Prices by Accommodation Category and Duration Bin

Category	7 Days & Under		8 Days & Over	
	Mean	SD	Mean	SD
Large All-Inclusive	1,216.55	381.61	1,538.03	626.17
Large European Plan	876.93	251.94	1,150.35	465.58
Medium European Plan	1,381.28	525.25	1,653.07	737.15
Small All-Inclusive	1,089.61	141.30	1,228.10	227.42
Small European Plan	864.02	215.90	982.78	320.76

All prices in 2024 USD; based on exit survey data 2001–2023.

Counterfactual Framework

- Using recovered demand parameters Θ , supply parameters Σ , and accounting parameters (employment ratios, outside-spending shares, tax rates), I solve for new equilibrium prices and quantities under alternative scenarios.
- **Development outcomes** evaluated in each counterfactual:
 - *Total tourist expenditures*: accommodation + outside-hotel spending.
 - *Employment*: workers per-tourist $\times$ arrivals (accommodation sector).
 - *Tax revenues*: GCT on accommodation sales + GART room-tax revenues.
 - *Consumer surplus / welfare* and *inside-option share* (arrivals to Jamaica).
- Four counterfactuals spanning market structure and tax policy.

Counterfactual 1: The “Spanish Invasion” of Montego Bay

- **Context:** Between 2006 and 2012, three Spanish-owned multinationals entered Montego Bay’s market: **Riu Hotels & Resorts**, **Iberostar Resorts**, and **Fiesta Group**.
- This roughly *doubled* the Large AI room capacity in the resort area.
- **Counterfactual scenarios:** I remove 1, 2, or all 3 of the new entrants, holding all other market and product characteristics constant.
- **Key question:** Did the multinationals attract new visitors to Jamaica, or did they divert demand from incumbents?

CF 1 Results: Montego Bay Entry Simulations

Table: Counterfactual 1: The ‘Spanish Invasion’ of Montego Bay (2006–2012)

Variable	Baseline (Levels)	Counterfactuals (%Δ)		
	Baseline	(1) Remove 1	(2) Remove 2	(3) Remove All
Development Outcomes				
Total Exp. (Millions USD)	838.65	-1.61	-3.29	-4.47
Total Tax Revenues (Millions USD)	48.06	-1.61	-3.29	-4.47
Employment	9477.00	-0.59	-1.11	-2.12
Demand Outcomes				
Average Price	1590.14	-1.07	-3.05	-2.80
Bookings (Thousands)	206.00	-0.60	-1.13	-2.09
HHI	2093.38	4.89	9.83	18.94

Notes: Columns 1–3 show average % change from baseline on removing 1, 2, or 3 entrants. Column 4 gives baseline levels. All monetary values in 2024 USD.

CF 1 Takeaways

- Removing the Spanish multinationals reduces **arrivals** substantially, confirming these brands attracted visitors who would otherwise not have come to Jamaica.
- The **diversion ratio** to the outside option for Large AI guests is high: incumbents see modest gains but not proportional to the capacity removed.
- **Employment** falls roughly in proportion to arrivals, since AI resorts employ more workers per room than EP hotels.
- **Tax revenues** fall with arrivals; concentration (HHI) rises when entrants are removed.
- *Bottom line:* Multinational AI entry was broadly beneficial for Montego Bay development outcomes.

Counterfactual 2: Negril Ownership Structure

- **Context:** 2006–2011, two incumbent parent companies (**Sandals** and **Couples Resorts**) each expanded their Negril-Southcoast portfolios by one property.
- Incumbents are multi-property firms; they internalize cross-property demand cannibalization, which dampens price competition.
- **Scenario 1:** Simulate the entrants operating as *single-property firms* (same quality, different ownership).
- **Scenario 2:** Simulate *no entry at all* — only original properties remain.
- **Key question:** Does multi-property ownership by incumbents suppress development-relevant outcomes relative to independent entry?

CF 2 Results: Negril Ownership Simulations

Table: Counterfactual 2: Negril Ownership Structure (2006–2011)

Variable	Baseline (Levels)	Counterfactuals (%Δ)	
	Baseline	(1) Ownership	(2) Removal
Development Outcomes			
Total Exp. (Millions USD)	910.01	-2.64	-9.63
Total Tax Revenues (Millions USD)	50.46	-2.64	-9.63
Employment	9281.00	2.22	-5.95
Demand Outcomes			
Average Price	1674.31	-4.64	-1.73
Bookings (Thousands)	213.00	2.22	-6.03
HHI	2256.50	-28.64	-3.05

Notes: Column 1 = single-property ownership counterfactual (% change from baseline). Column 2 = no entry (% change from baseline). Column 3 = baseline levels. All monetary values in 2024 USD.

CF 2 Takeaways

- Under single-property ownership, new entrants compete more aggressively on price → **lower prices, higher arrivals, more employment.**
- Multi-property incumbent expansion raises prices and dampens competitive pressure.
- No-entry scenario: arrivals and revenues fall sharply, confirming that entry per se is important even when incumbent ownership is the counterfactual.
- *Bottom line:* For development, the *structure* of entry matters; non-incumbent single-property entry outperforms multi-property expansion by incumbents.

Counterfactual 3: Room Tax (GART) Reform

- **Context:** Jamaica introduced the Guest Accommodation Room Tax (GART) in 2013 — a per-night, per-room specific tax with rates varying by property size. The *nominal* value has never been adjusted for inflation.
- **Scenario 1:** Complete **removal** of the GART.
- **Scenario 2: Inflation-index** the GART (i.e. update it to its 2024 real value).
- Sample covers 2014, 2018, and 2020–2023.
- **Key question:** Is the GART's erosion (in real terms) costing Jamaica revenue, and at what cost in terms of employment and arrivals?

CF 3 Results: Room Tax Simulations

Table: Counterfactual 3: GART Simulations by Panel

Variable	Baseline (Levels)	Counterfactuals (%Δ)	
	Baseline	(1) No GART	(2) Adj. GART
Development Outcomes			
Total Exp. (Millions USD)	3841.53	-0.69	-0.09
Total Tax Revenues (Millions USD)	230.18	-7.83	2.58
Employment	41645.00	0.25	-0.03
Demand Outcomes			
Average Price	1560.36	-0.90	-0.79
Bookings (Thousands)	926.00	0.24	-0.04
HHI	2375.25	0.05	0.03

Notes: Column 1 = GART removed. Column 2 = GART indexed to inflation. Column 3 = baseline levels. All monetary values in 2024 USD.

Counterfactual 4: GCT Tax Reform

- **Context:** Jamaica's General Consumption Tax (GCT) — a VAT-style tax — is charged at **10%** on all tourist accommodation. Same rate for luxury all-inclusives and small guesthouses.
- **Three scenarios** (covering 2012, 2014, 2017–2023):
 - ① **Tax cut for Small/Medium EP:** reduce GCT to 5%.
 - ② **Tax hike for Large AI and Large EP:** raise GCT to 15%.
 - ③ **Combined** (simultaneous cut + hike).
- Design motivated by the spending-spillover asymmetry between product types: small EP guests generate more local economic activity per tourist dollar.
- **Key question:** Can a more progressive GCT structure raise revenues without harming overall tourism?

CF 4 Results: GCT Simulations

Table: Counterfactual 4: GCT Simulations

Variable	Baseline (Levels)	Counterfactuals (%Δ)		
	Baseline	(1) Tax Cut	(2) Tax Increase	(3) Combined
Development Outcomes				
Total Exp. (Millions USD)	3865.57	-0.06	0.01	-0.05
Total Tax Revenues (Millions USD)	231.45	-8.45	32.64	24.11
Employment	42532.00	0.12	-0.21	-0.09
Demand Outcomes				
Average Price	1556.29	-0.25	1.11	0.87
Bookings (Thousands)	936.00	0.12	-0.20	-0.08
HHI	2398.91	-0.02	0.30	0.28

Notes: Col. 1 = GCT cut to 5% for Small/Medium EP. Col. 2 = GCT raised to 15% for Large AI and Large EP. Col. 3 = combined. Col. 4 = baseline levels. All monetary values in 2024 USD.

CF 4 Takeaways

- **Raising GCT on large AI/EP** increases tax revenues substantially, reflecting low price-sensitivity of their guests; modest impact on arrivals and employment.
- **Cutting GCT for small/medium EP** boosts their arrivals; raises employment and outside-hotel spending, but with a revenue loss.
- **Combined policy**: tax revenues rise by $\approx 55\%$ with minimal net impact on total tourist expenditures or aggregate employment.
- *Bottom line*: The flat 10% GCT is not optimal for development; a progressive structure exploits inelastic AI demand while supporting high-spillover small accommodations.

Future Simulation: Southcoast-Negril 2030 Expansion

Table: Counterfactual 5: Negril-Southcoast 2030 Expansion

Variable	Baseline (Levels)	Counterfactuals (%Δ)		
	Baseline	(1) New AIs	(2) New EPs	(3) AI and EPs
Development Outcomes				
Total Exp. (Millions USD)	1565.12	26.03	27.47	26.98
Total Tax Revenues (Millions USD)	93.40	27.71	27.47	27.40
Employment	12123.00	33.88	27.47	31.11
Demand Outcomes				
Average Price	1516.04	-5.53	0.00	6.75
Bookings (Thousands)	334.00	34.01	27.47	31.18
HHI	2171.21	-29.67	0.00	-17.21

Notes: Three scenarios for 2030 Negril-Southcoast expansion: (1) 2500 new AI rooms across 5 multi-owner properties, (2) 2500 new Small EP rooms, (3) mixed 1500 AI + 1000 Small EP. Baseline year: 2022. Market size = 600,000 parties.

Summary of Counterfactual Findings

	Arrivals	Employment	Tax Revenue	Exp.
CF1: Multi. AI entry (MoBay)	+	+	+	+
CF2: Non-incumbent entry (Neg)	+	+	+	+
CF3: Inflation-index GART	– small	– small	+	≈
CF4: Progressive GCT	≈	≈	++	≈

- Market structure matters: non-incumbent entry consistently dominates incumbent expansion.
- Tax reform can raise revenues substantially while being broadly neutral on employment.
- Price-inelastic AI demand is the key lever for revenue policy.

Conclusions

- 1 Jamaica's accommodations market features **price-inelastic demand**, especially for Large All-Inclusive resorts.
- 2 Large AI multinationals primarily bring **net new tourists** rather than diverting from incumbents, making them broadly beneficial for arrivals and employment.
- 3 However, their high within-nest inelasticity means they can **bear higher taxes** without large demand responses — useful for government revenue design.
- 4 A progressive GCT structure (5% for small EP, 15% for large AI/EP) raises revenues $\approx 55\%$ with minimal employment or expenditure impact.
- 5 Non-incumbent single-property entry **outperforms** multi-property expansion by incumbents for development outcomes.

Policy Implications

For Tax Policy

- Adopt a **progressive GCT structure** exploiting AI inelasticity.
- **Inflation-index the GART** to restore real revenue yield without triggering large demand losses.

For Competition and Investment Policy

- **Prioritize non-incumbent entry** when licensing new hotels; single-property ownership of new entrants generates stronger competitive discipline.
- The **Southcoast expansion** (currently being planned) would benefit from a mixed AI + small-EP product composition to maximize spillovers.

Thank you!

Comments and questions welcome.

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Demand: Utility Specification

A consumer i choosing booking b in market m receives utility

$$u_{ibm} = \alpha p_{bm} + \beta x_{bm} + \xi_{hm} + \eta_{inm} + (1 - \sigma_n)\epsilon_{ibm}.$$

- p_{bm} : per-person, per-booking price.
- x_{bm} : observed product characteristics.
- ξ_{hm} : unobserved mean utility common to all consumers.
- η_{inm} : nest-level idiosyncratic preference.
- $\sigma_n \in [0, 1)$: within-nest correlation for nest n .
- ϵ_{ibm} : iid Type-I extreme-value error.

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Demand: Nesting Structure

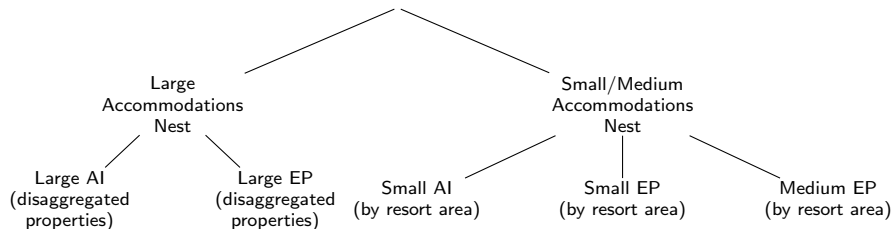


Figure: Nested structure of hotel categories. Duration bins (≤ 7 nights vs. ≥ 8 nights) are crossed with each product.

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Demand: Estimating Equation

The nested logit market-share equation is:

$$\ln(s_{bm}) - \ln(s_{0m}) = \beta x_{bm} + \alpha p_{bm} + \sigma_n \ln(\bar{s}_{bm|nm}) + \zeta_{bm},$$

where $\bar{s}_{bm|nm}$ is the within-nest share of booking b .

- Estimated by 2SLS; instruments address endogeneity of p_{bm} and $\bar{s}_{bm|nm}$.
- Market size M_m set as $2 \times$ total observed bookings, following the IO literature convention.

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Supply: Profit Function

Parent company f maximises:

$$\Pi_{fm}(p) = \sum_{h \in \mathcal{H}_f} \left[\sum_{d \in \mathcal{D}} \left(\frac{p_{dhm}}{1 + \tau^v} - c_{dhm} \right) q_{dhm} - \gamma_h g(\rho_{hm}; \lambda) \right],$$

where the congestion function is $g(\rho; \lambda) = \ln(1 + \frac{\rho}{\lambda(1-\rho)})$, and occupancy $\rho_{hm} = \frac{\sum_d \tilde{q}_{dhm} l_{dm}}{k_{hm}}$.

Linear marginal cost: $c_{dhm} \equiv \tilde{c}_{dhm} + \frac{\tau_{hm}^r l_{dm}}{l_{im}}$.

- τ^v : ad-valorem GCT.
- τ_{hm}^r : specific GART room tax (per night).
- γ_h : plan-specific congestion weight (AI vs. EP).

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Supply: FOC and Pricing Equation

Stacked FOC for Nash-Bertrand equilibrium:

$$\mathbf{q} + \mathbf{J}_q^\top [\Omega (\mathbf{p} - (1 + \tau^v) \mathbf{c})] - (1 + \tau^v) \mathbf{e} = \mathbf{0},$$

with $\Delta \equiv -(\Omega \circ \mathbf{J}_q)$ and $\mathbf{e}$ the congestion gradient vector. Rearranging:

$$\mathbf{p} = (1 + \tau^v) \mathbf{c} + \Delta^{-1} \mathbf{q} - (1 + \tau^v) \Delta^{-1} \mathbf{e}.$$

The markup vector is $\Delta^{-1} \mathbf{q}$, modified downward by the congestion-gradient correction $(1 + \tau^v) \Delta^{-1} \mathbf{e}$.

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Appendix: Instruments — First Stage

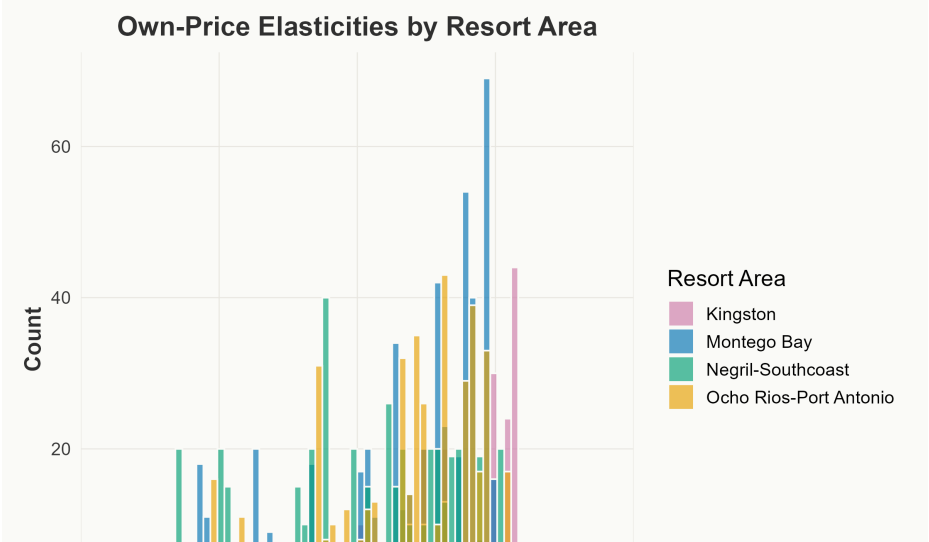
Statistics consistent for homoskedasticity only
Number of obs = 269

	deviation	Coefficient	Std. err.	t	P> t	[95% conf. interval]
LOG_SEARCH		-64050.77	29710.57	-2.16	0.032	-122601.6 -5499.925
product_ids						
112		10807.73	3461.082	3.12	0.002	3986.955 17628.51
113		-15872.21	3461.082	-4.59	0.000	-22692.99 -9051.428
114		-8571.44	3520.122	-2.43	0.016	-15508.57 -1634.311
121		-38079.04	3461.082	-11.00	0.000	-44899.82 -31258.26
122		-39189.35	3461.082	-11.32	0.000	-46010.13 -32368.57
123		-39099.3	3461.082	-11.30	0.000	-45920.09 -32278.52
124		-39173.97	3461.082	-11.32	0.000	-45994.75 -32353.19
221		-42460	3461.082	-12.27	0.000	-49280.78 -35639.22
222		-42576.58	3461.082	-12.30	0.000	-49397.36 -35755.8
223		-42883.22	3461.082	-12.39	0.000	-49704 -36062.44
224		-42742.61	3461.082	-12.35	0.000	-49563.39 -35921.83
321		-25329.46	3461.082	-7.32	0.000	-32150.24 -18508.68
322		-42493.87	3461.082	-12.28	0.000	-49314.65 -35673.09
323		-42691.7	3583.228	-11.91	0.000	-49753.19 -35630.21
324		-42411.82	3461.082	-12.25	0.000	-49232.6 -35591.04
product_ids#c.LOG_SEARCH						
112		28146.61	33078.13	0.85	0.396	-37040.7 93333.92
113		33521.8	33078.13	1.01	0.312	-31665.5 98709.11
114		32931.02	33095.1	1.00	0.321	-32289.74 98151.78
121		21963.57	33078.13	0.66	0.507	-43223.73 87150.88
122		29231.78	33078.13	0.88	0.378	-35955.53 94419.09
123		26906.67	33078.13	0.81	0.417	-38280.63 92093.98
124		24751.82	33078.13	0.75	0.455	-46435.49 89939.13
221		26098.6	33078.13	0.79	0.431	-39088.71 91285.91
222		27498.1	33078.13	0.83	0.407	-37689.21 92685.41
223		26817.26	33078.13	0.81	0.418	-38370.05 92004.56
224		26621.73	33078.13	0.80	0.422	-38565.58 91809.04
321		-20088.86	33078.13	-0.61	0.544	-85276.17 45098.45
322		22073.72	33078.13	0.67	0.505	-43113.59 87261.03
323		11354.44	35623.49	0.32	0.750	-58849.03 81557.91
324		18650.56	33078.13	0.56	0.573	-46536.75 83837.87

market_ids						
2005	6073.093	4292.027	1.60	0.111	-1585.236	15331.42
2006	2925.939	3917.54	0.75	0.456	-4794.387	10646.26
2008	3731.252	3582.996	1.04	0.299	-3329.705	10792.29
2009	3214.456	3312.764	0.97	0.333	-3314.032	9742.944
2010	356.8804	3462.823	0.10	0.918	-6467.331	7181.091
2011	-8799.48	3203.053	-2.75	0.007	-15111.76	-2487.201
2012	-11944.13	3799.73	-3.14	0.002	-19432.45	-4456.141
2013	-15351.66	4546.893	-3.38	0.001	-24312.24	-6391.008
2014	-14660.54	3337.152	-4.40	0.000	-21246.09	-8002.991
2015	-9325.001	2914.114	-3.20	0.002	-15068.67	-3582.935
2016	-6722.869	2908.056	-2.31	0.022	-12453.8	-991.9419
2017	-5191.322	3828.541	-1.36	0.176	-12736.26	2353.612
2018	-3565.307	3841.222	-0.93	0.354	-11135.23	4004.617
2019	-6192.465	4047.414	-1.53	0.127	-14168.73	1783.803
2020	0	(omitted)				
2021	0	(omitted)				
market_ids#c.LOG_SEARCH						
2005	0	(omitted)				
2006	0	(omitted)				
2008	0	(omitted)				
2009	0	(omitted)				
2010	0	(omitted)				
2011	0	(omitted)				
2012	0	(omitted)				
2013	0	(omitted)				
2014	0	(omitted)				
2015	0	(omitted)				
2016	0	(omitted)				
2017	0	(omitted)				
2018	0	(omitted)				
2019	0	(omitted)				
2020	0	(omitted)				
2021	0	(omitted)				
market_ids	711.4493	204.3564	3.48	0.001	308.7226	1114.176
_cons	-1385194	411409.1	-3.37	0.001	-2195961	-574427.1

Figure: First-stage instrument relevance (selected instruments)

Appendix: Resort Area Elasticities



Appendix: Category Diversion Ratios (Table)

Table: Mean Category Diversion Ratios

Category	Mean Diversion Ratios					
	Large AI Diversion	Large EP Diversion	Small AI Diversion	Medium EP Diversion	Small EP Diversion	Outside Option
Large All-Inclusive	0.482	0.011	0.012	0.005	0.050	0.440
Large European Plan	0.045	0.222	0.000	0.189	0.077	0.467
Medium European Plan	0.167	0.090	0.000	0.130	0.171	0.441
Small All-Inclusive	0.280	0.000	0.064	0.000	0.250	0.407
Small European Plan	0.263	0.044	0.035	0.113	0.117	0.429

Notes: Average diversion ratios conditional on a category not being available.

Appendix: Market Shares, Montego Bay

Table: Brand Market Shares — Montego Bay

Brand	Market Share		
	2001	2012	2023
Sunset	0.20	0.06	
Holiday Inn	0.17	0.10	0.05
Wyndham	0.12		
Small All-Inclusive Montego Bay	0.12	0.01	0.02
Sandals	0.10	0.11	0.08
Small European Plan Montego Bay	0.09	0.04	0.01
Trelawny Beach	0.08		
Medium European Plan Montego Bay	0.05	0.02	0.02
Grand Lido	0.04		
Ritz-Carlton	0.04	0.03	
Grand Palladium		0.16	0.08
Iberostar		0.13	0.08
Secrets		0.11	0.08
Divi		0.11	0.05

Appendix: HHI Across Years

Table: Herfindahl-Hirschman Index (HHI) by Year

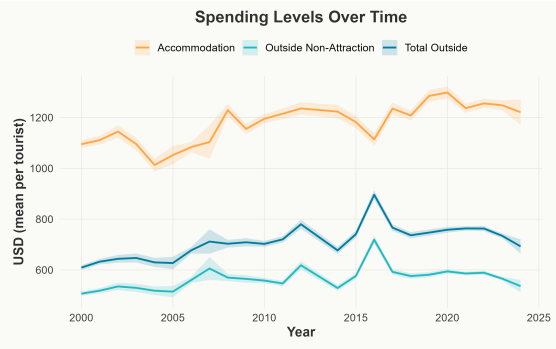
Resort Area	HHI Type	2001	2012	2023
Montego Bay	Booking HHI	1,091.00	940.00	992.00
	Room Stock HHI	356.00	483.00	867.00
	Room Nights HHI	1,194.00	940.00	979.00
Ocho Rios	Booking HHI	1,841.00	1,306.00	1,702.00
	Room Stock HHI	451.00	524.00	1,404.00
	Room Nights HHI	1,653.00	1,332.00	1,760.00
Negril	Booking HHI	709.00	1,157.00	981.00
	Room Stock HHI	205.00	359.00	680.00
	Room Nights HHI	583.00	1,106.00	896.00
Kingston	Booking HHI	1,422.00	1,051.00	733.00
	Room Stock HHI	721.00	594.00	911.00
	Room Nights HHI	1,229.00	946.00	724.00

Appendix: Capacity by Category and Year

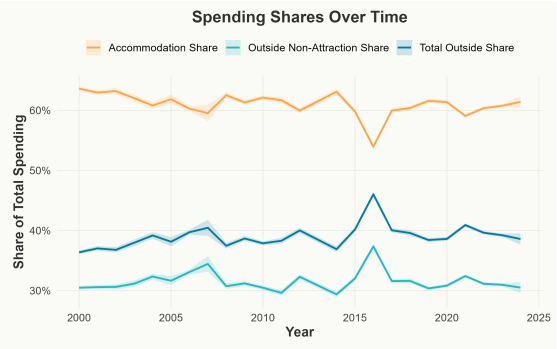
Table: Room-Night Capacity by Accommodation Category

Year	Category	Room Capacity	Unique Properties	Room Nights Capacity
2001	Large All-Inclusive	12,056.00	40.00	4,216,328.00
	Large European Plan	2,696.00	6.00	984,040.00
	Medium European Plan	592.00	14.00	627,600.00
	Small All-Inclusive	272.00	14.00	441,650.00
	Small European Plan	23,670.00	3,480.00	8,639,550.00
2011	Large All-Inclusive	26,434.00	60.00	9,565,242.00
	Large European Plan	1,236.00	4.00	451,140.00
	Medium European Plan	286.00	8.00	359,160.00
	Small All-Inclusive	272.00	12.00	364,270.00
	Small European Plan	23,110.00	3,428.00	8,435,150.00
2021	Large All-Inclusive	29,280.00	56.00	10,687,200.00
	Large European Plan	1,068.00	4.00	389,820.00
	Medium European Plan	526.00	12.00	554,070.00
	Small All-Inclusive	192.00	8.00	219,000.00
	Small European Plan	15,750.00	1,228.00	5,748,750.00

Appendix: Outside Spending Over Time



(a) Average Outside-Hotel Spending Levels



(b) Average Outside-Hotel Spending Shares

Appendix: Large AI vs. Small EP — Spending Comparison

Table: Outside Spending: Large AI vs. Small EP

Metric	Large All-Inclusive			Small European Plan			Difference & Test (LAI – SEP)	
	N (LAI)	Mean (LAI)	SD (LAI)	N (SEP)	Mean (SEP)	SD (SEP)	LAI – SEP	p-value
Outside Non-Attraction Share	86910	0.31	0.12	8279	0.39	0.19	-0.08	¡0.0001
Total Outside Exp. Share	86910	0.39	0.12	8279	0.43	0.20	-0.04	¡0.0001
Accommodation Share	86910	0.61	0.12	8279	0.57	0.20	0.04	¡0.0001
Outside Non-Attraction Spending	86910	567.45	350.61	8279	619.62	536.63	-52.18	¡0.0001
Total Outside Spending	86910	733.97	419.76	8279	691.34	569.80	42.63	¡0.0001
Accommodation Spending	86910	1202.57	710.71	8279	998.40	859.56	204.17	¡0.0001

Notes: Based on exit survey data.